

TP3 filtering

Load the filter coefficients a_k and b_k :

```
coef=np.loadtxt("coefbpcours.csv",delimiter=',')
```

```
b=coef[0,:] # non recursif coef
```

```
a=coef[1,:] # recursif coef
```

Exercise 1:

Compute the impulse response $h(n)$.

Compute the frequency response. What kind of filter is it ?

What can you say about the filter stability ?

By using the function $z,p,k=\text{scipy.signal.tf2zpk}(b,a)$, plot the poles and the zeros in the complex plane. States the filter stability.

Exercise 2:

Check the frequency response by using a white noise.

Exercise 3:

Plot the frequency response by directly using the a_k and b_k

Exercise 4:

Generate a sweeping frequency signal (from 0 to 0.5) . filter it and draw the temporal response.

Exercise 5 :

Take the z,p,k from exercise 1 and create a new set of a_k and b_k : $\text{newb,newa}=\text{sci.zpk2tf}(z,1/p, k)$

plot the frequency response of this new filter in the same way as exercise 3 : conclusions ?

plot the pole and zeros as in exercise 1. conclusions ?

Plot the impulse response